

# UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

## Functions of several variables: Taylor Formula

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### 0.1 Higher order derivatives

Assume that  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  is differentiable. One can ask whether the new functions  $f_x$  and  $f_y$  (the partial derivatives) are continuous or even also differentiable. The following theorem on the continuity of partial derivatives is often used as a sufficient condition for the differentiability.

#### Theorem 0.1 (A sufficient condition for differentiability)

Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be such that both partial derivatives  $f_x(x, y)$  and  $f_y(x, y)$  exist at every  $(x, y)$  in a small disc centered at the point  $(a, b)$  and are continuous at the point  $(a, b)$ . Then the function  $f$  is differentiable at  $(a, b)$ .

We omit the proof in these notes. Notice that a function can be differentiable at a point without its partial derivative being continuous at that point. This is a well-known fact even for functions of a single variable. For instance, the function

$$f(x) := \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable at  $x = 0$  but its derivative

$$f'(x) := \begin{cases} 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) & \text{if } x \neq 0, \\ 0 & \text{otherwise} \end{cases}$$

is not continuous at  $x = 0$ . **(Check that this is indeed its derivative and that  $f'$  is not continuous at  $x = 0$ !)**

The study of the differentiation of the new functions  $f_x$  and  $f_y$  leads to the notion of successive differentiation. For instance, we will have to find the second order partial derivatives

$$\begin{aligned} f_{xx} &= \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial x} \right), & f_{xy} &= \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right), \\ f_{yx} &= \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right), & f_{yy} &= \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial y} \right) \end{aligned}.$$

Let us first observe that  $f_{xy}$  and  $f_{yx}$  do not always coincide as one can see from the function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined by

$$f(x, y) := \begin{cases} y^2 \arctan\left(\frac{x}{y}\right) & \text{if } y \neq 0, \\ 0 & \text{if } y = 0. \end{cases}$$

for which one can find that

$$f_x(x, y) = \begin{cases} \frac{y}{1 + \frac{x^2}{y^2}} & \text{if } y \neq 0, \\ 0 & \text{if } y = 0, \end{cases} \quad \text{while} \quad f_y(x, y) = \begin{cases} 2y \arctan\left(\frac{x}{y}\right) - \frac{x}{1 + \frac{x^2}{y^2}} & \text{if } y \neq 0, \\ 0 & \text{if } y = 0. \end{cases}$$

$$\begin{cases} f_{xy}(0, 0) = \frac{\partial f_y}{\partial x}(0, 0) := \lim_{x \rightarrow 0} \frac{f_y(x, 0) - f_y(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0 \\ f_{yx}(0, 0) = \frac{\partial f_x}{\partial y}(0, 0) := \lim_{y \rightarrow 0} \frac{f_x(0, y) - f_x(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{y - 0}{y} = 1 \end{cases}$$

Therefore, as one can see that

$$f_{xy}(0, 0) = 1 \neq 0 = f_{yx}(0, 0).$$

The following theorem gives us a sufficient condition under which  $f_{xy}$  and  $f_{yx}$  coincide.

**Theorem 0.2 (Clairaut-Schwartz)**

Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ . Assume that the mixed partial derivatives  $f_{xy}$  and  $f_{yx}$  exist on a disc  $D$  centered at the point  $(a, b)$  and are continuous at that point. Then

$$f_{xy}(a, b) = f_{yx}(a, b). \quad (1)$$

We omit the proof in this course!

We defined above functions of class  $C^1$  as functions having their first order partial derivatives continuous. In general, we have the following definition.

**Definition 0.3** A function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  is said to be of class  $\mathcal{C}^2$  if all its partial derivatives up to order 2 are continuous.

In general, a function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  is said to be of class  $\mathcal{C}^k$  ( $k \geq 3$ ) if all its partial derivatives up to order  $k$  are continuous.

If the function  $f$  is of class  $\mathcal{C}^2$ , then the matrix

$$Hf(x, y) := \begin{pmatrix} f_{xx}(x, y) & f_{yx}(x, y) \\ f_{xy}(x, y) & f_{yy}(x, y) \end{pmatrix}$$

is called the *Hessian matrix* of  $f$  at the point  $(x, y)$ .

For functions of three variables the Hessian matrix of  $f$  at the point  $(x, y, z)$  is

$$Hf(x, y, z) := \begin{pmatrix} f_{xx}(x, y, z) & f_{yx}(x, y, z) & f_{zx}(x, y, z) \\ f_{xy}(x, y, z) & f_{yy}(x, y, z) & f_{zy}(x, y, z) \\ f_{xz}(x, y, z) & f_{yz}(x, y, z) & f_{zz}(x, y, z) \end{pmatrix}$$

It follows from Theorem 0.2 that the Hessian matrix  $Hf(x, y)$  or  $Hf(x, y, z)$  of the function  $f$  of class  $\mathcal{C}^2$  at the point  $(x, y)$  or  $(x, y, z)$  is symmetric.

## 0.2 The Taylor formula for functions of several variables

In this section, we will present a brief account of Taylor formula for functions of several variables. We will deal here with functions of class  $\mathcal{C}^2$  i.e., functions with continuous partial derivatives up to order 2. Let us first recall the Taylor formula for functions of a single variable, up to order 2. In order, to make the presentation lighter, we opted for functions defined on the whole line  $\mathbb{R}$ .

**Theorem 0.4** *Let  $t_0 \in \mathbb{R}$ . Let  $\phi : \mathbb{R} \rightarrow \mathbb{R}$  be continuously differentiable with the second derivative which exists everywhere except possibly at  $t_0$ . Then for every  $t \in \mathbb{R}$ , there exists  $s$  between  $t$  and  $t_0$  such that*

$$\phi(t) = \phi(t_0) + \frac{1}{1!}\phi'(t_0)(t - t_0) + \frac{1}{2!}\phi''(s)(t - t_0)^2. \quad (2)$$

The identity (0.4) is called **the second order Taylor formula with the Lagrange remainder**.

Now, let  $(a, b) \in \mathbb{R}^2$  and let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function of class  $\mathcal{C}^2$ . Let  $(x, y) \in \mathbb{R}^2$ . We set

$$\phi(t) := f((a, b) + t(x - a, y - b)) \quad (3)$$

$$\phi(0) = f(a, b), \quad \phi(1) = f(x, y), \quad \phi'(t) = \nabla f((a, b) + t(x - a, y - b)) \cdot (x - a, y - b) \quad (\text{CR!!!}).$$

We write

$$\phi'(t) = f_x((a, b) + t(x - a, y - b))(x - a) + f_y((a, b) + t(x - a, y - b))(y - b).$$

Thus, from the Chain Rule applied twice, we get that

$$\begin{aligned} \phi''(t) &= \nabla f_x((a, b) + t(x - a, y - b)) \cdot (x - a, y - b)(x - a) \quad (\text{CR}) \\ &\quad + \nabla f_y((a, b) + t(x - a, y - b)) \cdot (x - a, y - b)(y - b) \quad (\text{CR}) \\ &= f_{xx}((a, b) + t(x - a, y - b))(x - a)^2 + f_{yx}((a, b) + t(x - a, y - b))(x - a)(y - b) \\ &\quad + f_{xy}((a, b) + t(x - a, y - b))(x - a)(y - b) + f_{yy}((a, b) + t(x - a, y - b))(y - b)^2 \\ &= f_{xx}((a, b) + t(x - a, y - b))(x - a)^2 + 2f_{xy}((a, b) + t(x - a, y - b))(x - a)(y - b) \\ &\quad + f_{yy}((a, b) + t(x - a, y - b))(y - b)^2 \quad (\text{because } f_{xy} = f_{yx}) \\ &= \left( (x - a, y - b) \cdot Hf((a, b) + t(x - a, y - b)) \begin{pmatrix} x - a \\ y - b \end{pmatrix} \right). \end{aligned}$$

Now, when we apply the Taylor formula (2) to the function  $\phi$ , we get for some  $s \in (0, 1)$  that

$$\phi(1) = \phi(0) + \frac{1}{1!}\phi'(0)(1 - 0) + \frac{1}{2!}\phi''(s)(1 - 0)^2.$$

That is,

$$f(x, y) = f(a, b) + \frac{1}{1!}\nabla f(a, b) \cdot (x - a, y - b) + \frac{1}{2!}(x - a, y - b) \cdot Hf(\xi, \eta) \begin{pmatrix} x - a \\ y - b \end{pmatrix} \quad (4)$$

with  $(\xi, \eta) = (a, b) + s(x - a, y - b)$  and  $s \in (0, 1)$ . Such a point  $(\xi, \eta)$  lies on the segment line joining the points  $(a, b)$  to  $(x, y)$ .

The identity (4) is called the **second order Taylor formula for the function  $f$  with the Lagrange remainder**.

We will need (4) later for the study of unconstrained optimization for functions of two variables. Specially, when we will be examining (as in the case of single variable functions studied in Calculus I) some second order conditions in order to determine whether a critical point for some given function is a point of local maximum/minimum or neither.

**The next set of notes will be on parametric surfaces I!**